Connections between Fields and Particles.

Problems: Non-unitary representation of Poincaré group
(Lorentz group)

example: we've learnt that the spinors transform under Lorentz group like

$$\psi_R \to e^{\frac{1}{2}\sum_j (i\theta_j \sigma_j + \beta_j \sigma_j)} \psi_R$$

$\uparrow$ rotation with rotation generator $J_j = \frac{i}{2}\sigma_j$ $\quad$ $\uparrow$ Boost.

Obviously, the spinor representation isn't unitary $\Lambda_{1/2}^{\dagger}\Lambda_{1/2} \neq 1$ in Boost part (this is because Lorentz group is noncompact) that is, β is unbounded.

The fact that Lorentz group is noncompact means to have unitary representation, we must have infinite dimensional representation.

But we know, fields as functions of space-time transform under Lorentz transformations as finite-dimensional representations of the Lorentz group. (Poincaré group)

What constrains should be imposed when we promote fields into operators?

$$\Rightarrow U(\Lambda, a)\, \phi_\ell(x)\, U^{-1}(\Lambda, a) = \sum_{\bar{\ell}} D_{\ell\bar{\ell}}(\Lambda^{-1})\, \phi_{\bar{\ell}}(\Lambda x + a)$$

example: $U(\Lambda, a)\, V^\mu(x)\, U^{-1}(\Lambda, a) = (\Lambda^{-1})^\mu{}_\nu V^\nu(\Lambda x + a)$

vector field

$\updownarrow$ different!

$$V^\mu(x) \to V'^\mu(x') = \Lambda^\mu{}_\nu V^\nu(x)$$

This can be derived by consider position operator and rotation

$X_i |x\rangle = x_i |x\rangle$

$U_R X_i |x\rangle = U_R X_i U_R^{-1} U_R |x\rangle = U_R X_i U_R^{-1} |Rx\rangle$

$= x_i |Rx\rangle = U_R X_i U_R^{-1} |x'\rangle \quad \Rightarrow U_R X_i U_R^{-1} = (R^{-1}X)_i$

$= (R^{-1}x')_i |x'\rangle$ define $x' = Rx$

Thus we got a strong constain $U(\Lambda, a)\, \phi_\ell(x)\, U^{-1}(\Lambda, a) = \sum_{\bar{\ell}} D_{\ell\bar{\ell}}(\Lambda)\, \phi_{\bar{\ell}}(\Lambda x + a)$

this will tell us the form of coefficient if we do mode expansion.

by regarding fields as creation and annililation of particles

(embed particles into fields)

$$\phi_\ell^+(x) = \sum_\sigma \int \frac{d^3p}{(2\pi)^3 \sqrt{2p^0}}\, u_\ell(\sigma, \mathbf{p})\, a(\mathbf{p}, \sigma)\, e^{-ipx}$$

$$\phi_\ell^-(x) = \sum_\sigma \int \frac{d^3p}{(2\pi)^3 \sqrt{2p^0}}\, v_\ell(\sigma, \mathbf{p})\, a^\dagger(\mathbf{p}, \sigma)\, e^{+ipx}$$

One-particle state: Lorentz group has infinite-dimensional unitary representation on one-particle state $|p, \sigma\rangle$

$\uparrow$ other quantum number

$\uparrow$ 4-momentum

$$U(\Lambda) |p, \sigma\rangle = \sum_{\sigma'} C_{\sigma\sigma'}(\Lambda, p)\, |\Lambda p, \sigma'\rangle$$

By method of Little group we define a standard momentum k.

and any one-particle-state with momentum p is defined through

$|p, \sigma\rangle \equiv U(L(p;k)) |k, \sigma\rangle$, where L is Lorentz transform

brings k to p

we can see that $U(\Lambda)|p,\sigma\rangle = U(\Lambda)U(L(p;k))|k,\sigma\rangle$

$$= U(L(\Lambda p;k))U(L^{-1}(\Lambda p;k)\Lambda L(p;k))|k,\sigma\rangle$$

$W(\Lambda,p,k) \equiv L^{-1}(\Lambda p;k)\Lambda L(p;k)$

↳ form a subgroup that keeps k invariant.

↳ Little group.

Finding the representation of little group is much easier.

$$U(W(\Lambda,p,k))|k,\sigma\rangle = \sum_{\sigma'} D_{\sigma\sigma'}(W(\Lambda,p,k))|k,\sigma'\rangle$$

$$\Rightarrow U(\Lambda)|p,\sigma\rangle = \sum_{\sigma'} D_{\sigma\sigma'}(W(\Lambda,p,k))|\Lambda p,\sigma'\rangle$$

example: take $k=(m,0,0,0)$ for massive particle.

little group would be $SO(3)$ group.

If we define $\langle p';\sigma'|p,\sigma\rangle \equiv \delta_{\sigma\sigma'}\delta^{(3)}(\mathbf{p}'-\mathbf{p})$

but this is actually not Lorentz invariant $\langle p',\sigma'|U^{-1}(L(p;k))U(L(p;k))|p,\sigma\rangle$

$$= \langle k',\sigma'|k,\sigma\rangle$$
$$= \delta_{\sigma\sigma'}\delta^{(3)}(\mathbf{k}'-\mathbf{k})$$

Actually there should be normalization coefficient

$$|p,\sigma\rangle \equiv N(p;k)\,U(L(p,k))|k,\sigma\rangle$$

$$\Rightarrow \delta_{\sigma\sigma'}\delta^{(3)}(\mathbf{p}'-\mathbf{p}) = |N(p;k)|^2\delta_{\sigma\sigma'}\delta^{(3)}(\mathbf{k}'-\mathbf{k})$$

$$\int d^4p\,\delta(p^2-M^2)\,\theta(p^0)\,f(p) = \int d^3p\,dp^0\,\delta(\underbrace{p^{0\,2}-|\mathbf{p}|^2-M^2})\,\theta(p^0)\,f(p) = \int d^3p\,\frac{f(\mathbf{p},\sqrt{|\mathbf{p}|^2+M^2})}{2\sqrt{|\mathbf{p}|^2+M^2}}$$

integrate over p^0

$$\int dq^0\,\delta(f(q^0)) = \int \frac{dq^0}{|f'(q^0)|}\delta(q^0)$$

$$\left[(p^0-\sqrt{|\mathbf{p}|^2+M^2})(p^0+\sqrt{|\mathbf{p}|^2+M^2})\right]'$$
$$= p^0+\sqrt{|\mathbf{p}|^2+M^2}+p^0-\sqrt{|\mathbf{p}|^2+M^2} = 2p^0$$

$\Rightarrow \delta_{\sigma\sigma'} p^0 \delta^{(3)}(\mathbf{p}'-\mathbf{p}) = \delta_{\sigma\sigma'} k^0 \delta^{(3)}(\mathbf{k}'-\mathbf{k})$ is Lorentz invariant.

$\Rightarrow N(p;k) = \sqrt{\frac{k^0}{p^0}}$

$\therefore |p,\sigma\rangle = \sqrt{\frac{k^0}{p^0}}\, U(L(p;k))\, |k,\sigma\rangle$

Creation operator acting on vaccum state.

$|p,\sigma\rangle \equiv \sqrt{2p^0}\, a^\dagger(\mathbf{p},\sigma)|0\rangle \Rightarrow \langle p',\sigma'|p,\sigma\rangle = 2p^0 \langle 0| a(\mathbf{p}',\sigma')\, a^\dagger(\mathbf{p},\sigma)|0\rangle$

$\because [a(\mathbf{p}',\sigma'), a^\dagger(\mathbf{p},\sigma)] = \delta_{\sigma\sigma'}\, \delta^{(3)}(\mathbf{p}'-\mathbf{p})$

$$\Rightarrow U(\Lambda,a)|p,\sigma\rangle = e^{-i\Lambda p\cdot a}\sqrt{\frac{p_\Lambda^0}{p^0}} \sum_{\sigma'} D_{\sigma\sigma'}(W(\Lambda,p,k))\, \underbrace{|\Lambda p,\sigma'\rangle}_{\sqrt{2p_\Lambda^0}\, a^\dagger(\mathbf{p}_\Lambda,\sigma')|0\rangle}$$

$$= \sqrt{2p^0}\, U(\Lambda,a)\, a^\dagger(\mathbf{p},\sigma)\, U^{-1}(\Lambda,a)\, |0\rangle$$

$$U(\Lambda,a)\, a^\dagger(\mathbf{p},\sigma)\, U^{-1}(\Lambda,a) = \frac{p_\Lambda^0}{p^0}\, e^{-i\Lambda p\cdot a} \sum_{\sigma'} D_{\sigma\sigma'}(W(\Lambda,p,k))\, a^\dagger(\mathbf{p}_\Lambda,\sigma')$$

$$U(\Lambda,a)\, \phi_\ell^-(x)\, U^{-1}(\Lambda,a) = \sum_{\bar\ell} D_{\ell\bar\ell}(\Lambda^{-1})\, \phi_{\bar\ell}^-(\Lambda x+a)$$

$$\Rightarrow \sum_\sigma \int \frac{d^3p}{(2\pi)^3\sqrt{2p^0}}\, v_\ell(\sigma,\mathbf{p})\, \frac{p_\Lambda^0}{p^0}\, e^{-i\Lambda p\cdot a} \sum_{\sigma'} D_{\sigma\sigma'}(W)\, a^\dagger(\mathbf{p}_\Lambda,\sigma')\, e^{-ip\cdot x}$$

$$= \sum_{\bar\ell} D_{\ell\bar\ell}(\Lambda^{-1}) \sum_\sigma \int \frac{d^3p}{(2\pi)^3\sqrt{2p^0}}\, v_{\bar\ell}(\sigma,\mathbf{p})\, a^\dagger(\mathbf{p},\sigma)\, e^{-ip(\Lambda x+a)} \quad \text{let } p=\Lambda q$$

$$\sum_\sigma \sum_{\sigma'} v_\ell(\sigma,\mathbf{p})\, D_{\sigma\sigma'}(W) = \sqrt{\frac{p^0}{p_\Lambda^0}} \sum_{\bar\ell} \sum_\sigma v_{\bar\ell}(\sigma,\mathbf{p}_\Lambda)\, D_{\ell\bar\ell}(\Lambda^{-1})$$